

Why Data Cleaning Matters: Methods and Tools in R

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This handout introduces the concept and practice of data cleaning as a fundamental step in preparing datasets for reliable analysis. The process of data cleansing is integral to the maintenance of data accuracy, consistency, and integrity. It addresses common issues such as duplicates, missing values, and inconsistent formats. The document emphasises the key benefits of data cleaning, including improved analytical validity, enhanced comparability across datasets, and stronger predictive modelling outcomes. A practical business use case is provided to illustrate how survey data can be cleaned to produce trustworthy insights for marketing decisions. In addition, the handout demonstrates how the tidyverse collection of R packages, focusing on `dyplr`, offers powerful functions for structuring and standardizing data. By combining theoretical explanations with hands-on coding examples, the material equips students with both conceptual understanding and practical skills for transforming raw information into clean, analysis-ready datasets.

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Word count: 701

1 Data Cleaning

1.1 Key Benefits of Data Cleaning:

- **Accuracy** of results by removing errors, duplicates or other inconsistencies
- **Consistency** across datasets makes it possible to merge and compare datasets without conflict
- Handling **missing values** to prevent biased outcomes
- Better modeling and **predictions** when using statistical models

see Kabacoff (2024), Lee (2025)

1.2 Business Use Case

Scenario: A company runs an online survey to measure customer satisfaction after a product launch.

Problem: Responses include *missing demographic data*, *inconsistent gender categories* (“M”, “Male”, “m”), and some people *submit multiple times*

Data Cleaning Steps:

- Remove duplicate responses to avoid bias
- Standardize categories (e.g., all “M” -> “Male”)
- Handle missing values by either removing incomplete responses or filling defaults

Business Impact:

- Survey results accurately reflect customer opinions
- Marketing teams can segment customers by demographics with confidence
- Decisions about product improvements or new campaigns are based on reliable insights

2 Data Cleaning in R

2.1 Overview of Functions

tbl. 1.: Selection of Functions in Tidyverse

Package	Function	Use
dplyr	select()	select variables/columns
dplyr	filter()	select observations/rows
dplyr	mutate()	transform or recode variables
dplyr	summarize()	summarize data
dplyr	group_by()	identify subgroups for further processing
dplyr	distinct()	remove duplicate rows to ensure uniqueness
dplyr	arrange()	order rows by one or more variables (e.g., sort by age)
tidyr	drop_na()	remove rows with missing values in selected columns
tidyr	replace_na()	replace missing values with default (e.g., “none”)
tidyr	pivot_longer()	reshape data from long to wide format
tidyr	pivot_wider()	reshape data from long to wide format
tibble	tribble()	construct small tibbles by hand

As shown in *Table 1*, different packages provide specialized tools to handle messy datasets in R. The **tidyverse** is a collection of such packages, including *dplyr*, *tidyr*, *stringr*, *forcats*, and *janitor*. As mentioned in Kabacoff (2024) and Wickham (2023), each package offers functions to select, filter, reshape, and standardize data. Combining them makes it easier to transform raw information into clean and consistent datasets.

2.2 Data Cleaning Using dplyr

To reproduce simply the following coding exercise using some of the functions presented in *Table 1*, one may utilize the **starwars** dataset, which is included in the tidyverse collection. Access to this dataset requires loading the tidyverse package before starting this exercise, thereby ensuring that the data and relevant functions are available for practice.

2.2.1 select()

- keeps only the columns you need, removes irrelevant ones

```
starwars %>%  
select(name, species, contains("color")) %>%
```

2.2.2 filter()

- keeps only rows that meet certain conditions

```
starwars %>%  
  select(name, height, species, eye_color) %>%  
  filter(species == "Human" &  
         height < 200 &  
         eye_color %in% c("blue", "brown")) %>%
```

2.2.3 mutate()

- creates new variables or modifies existing ones

```
starwars %>%  
  select(name, height, mass, species) %>%  
  mutate(height = height/100) %>%
```

2.2.4 mutate()

- creates new variables or modifies existing ones

```
starwars %>%  
  select(name, height, mass, species) %>%  
  mutate(height_2 = height/100) %>%
```

2.2.5 mutate() and recode ()

- replace or rename categories for consistency

```
starwars %>%  
  select(name, contains("color"), species) %>%  
  mutate(species = recode(species,  
                          "Droid" = "Robot")) %>%
```

2.2.6 missing data - na.omit()

- drops all rows with missing values

```
starwars %>%  
  select(name, gender, hair_color, height) %>%  
  na.omit()
```

2.2.7 missing data - filter()

- shows all rows with missing value

```
starwars %>%  
  select(name, gender, hair_color, height) %>%  
  filter(!complete.cases(.))
```

2.2.8 missing data - drop.na()

- remove rows only if specific columns have missing values

```
starwars %>%
  select(name, gender, hair_color, height) %>%
  filter(!complete.cases(.)) %>%
  drop_na(height)
```

2.2.9 missing data - replace_na()

- fills missing values with defaults

```
starwars %>%
  select(name, gender, hair_color, height) %>%
  filter(!complete.cases(.)) %>%
  mutate(hair_color = replace_na(hair_color, "none"))
```

2.2.10 group_by () and summarise ()

- organise data into groups (grouped operations) and collapse groups into summary statistics (e.g., mean, count, sum)

```
starwars %>%
  group_by(species) %>%
  summarise(avg_mass = mean(mass, na.rm = TRUE))
```

3 Recommendations

For further insights into the topic of data cleaning and its applications, students may consult Kabacoff (2024), Wickham (2023) and Nordmann et al. (2022). These sources provide extended discussions and examples that complement the material presented in this hand-out. Individuals who favour teaching videos and experiential learning activities may find the Youtube channel “R Programming 101”’s R_Programming_101 (2025) tutorials and hands-on exercises particularly beneficial.

4 References

- Kabacoff, Robert. 2024. *Modern Data Visualization with r*. Chapman; Hall/CRC.
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- R_Programming_101. 2025. “R Programming 101.” <https://www.youtube.com/@RProgramming101>.
- Wickham, Mine; Grommund, Hadley; Centinkaya-Rundel. 2023. *R for Data Science: Import, Tidy, Transform, Visualize, and Model Data*. O’Reilly Media, Inc.

5 Affidavit

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